SHEAFIFICATION OF AN ABELIAN PRESHEAF CARRIES A UNIQUE ABELIAN SHEAF STRUCTURE AND IS LEFT ADJOINT

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Allowed standard prerequisites (editorial).

- Lemma lemma-stalk-abelian-presheaf (canonical tag 007F): stalks of abelian presheaves are abelian groups and the stalk functor is compatible with the algebraic structure. cited at proof lines 1955-1957 and 1977-1979. Gives the abelian group structure on stalks used throughout.
- Example example-sheaf-product-pointwise-algebraic-structure (canonical tag 007R): pointwise algebraic structures on products of stalks give a sheaf of algebraic structures. cited at proof lines 1957-1959. Makes Pi(F) a sheaf of abelian groups.
- Lemma lemma-diagram-fibre-product (canonical tag 0084): the sheafification value F^#(U) is the fibre product of a diagram of the stalks. cited at proof lines 1969-1972. Identifies F^#(U) as a fibre product of abelian groups, hence a subgroup of Pi(F)(U).
- Lemma lemma-stalk-sheafification (canonical tag 007Z): the stalk of the sheafification is canonically identified with the stalk of the presheaf. cited at proof lines 1979-1981 and 1989-1990. Identifies F_x with F^#_x as abelian groups and lets the abelian property be checked on stalks.
- Lemma lemma-check-homomorphism-stalks (canonical tag 007W): a map of sheaves is a homomorphism of algebraic structures iff it is on all stalks. cited at proof lines 1990-1991. Upgrades the set-level adjoint morphism to a morphism of abelian sheaves.

Source reference. Original lemma 007F: sources/stacks-chow/tags.txt, source lines 279–279. Cited at proof lines 1955-1957 and 1977-1979; load-bearing inside the counted span.

Source reference. Original remark 007R: sources/stacks-chow/tags.txt, source lines 291–291. Cited at proof lines 1957-1959 to make Pi(F) a sheaf of abelian groups; load-bearing inside the counted span (Example cited as kind=remark per kind mapping).

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Source reference. Original section 007X: sources/stacks-chow/tags.txt, source lines 297–297. Cited at proof lines 1953-1954 and 1962-1963 for the construction being used; navigation for the setting.

Source reference. Original lemma 0084: sources/stacks-chow/tags.txt, source lines 304–304. Cited at proof lines 1969-1972 for the subgroup argument; load-bearing inside the counted span.

Source reference. Original lemma 007Z: sources/stacks-chow/tags.txt, source lines 299–299. Cited at proof lines 1979-1981 and 1989-1990; load-bearing inside the counted span.

Source reference. Original lemma 007W: sources/stacks-chow/tags.txt, source lines 296–296. Cited at proof lines 1990-1991 to upgrade the adjoint to abelian sheaves; load-bearing inside the counted span.

Lemma 0.2. *Let X be a topological space. Let $\mathcal{F}$ be an abelian presheaf on X. Then there exists a unique structure of abelian sheaf on $\mathcal{F}^\#$ such that $\mathcal{F} \to \mathcal{F}^\#$ is a morphism of abelian presheaves. Moreover, the following adjointness property holds*

$$\mathrm{Mor}_{PAb(X)}(\mathcal{F}, i(\mathcal{G})) = \mathrm{Mor}_{Ab(X)}(\mathcal{F}^\#, \mathcal{G}).$$

Proof. Recall the sheaf of sets $\Pi(\mathcal{F})$ defined in Section 007X. All the stalks $\mathcal{F}_x$ are abelian groups, see Lemma 007F. Hence $\Pi(\mathcal{F})$ is a sheaf of abelian groups by Example 007R. Also, it is clear that the map $\mathcal{F} \to \Pi(\mathcal{F})$ is a morphism of abelian presheaves. If we show that condition $(*)$ of Section 007X defines a subgroup of $\Pi(\mathcal{F})(U)$ for all open subsets $U \subset X$, then $\mathcal{F}^\#$ canonically inherits the structure of abelian sheaf. This is quite easy to do by hand, and we leave it to the reader to find a good simple argument. The argument we use here, which generalizes to presheaves of algebraic structures is the following: Lemma 0084 show that $\mathcal{F}^\#(U)$ is the fibre product of a diagram of abelian groups. Thus $\mathcal{F}^\#$ is an abelian subgroup as desired.

Note that at this point $\mathcal{F}_x^\#$ is an abelian group by Lemma 007F and that $\mathcal{F}_x \to \mathcal{F}_x^\#$ is a bijection (Lemma 007Z) and a homomorphism of abelian groups. Hence $\mathcal{F}_x \to \mathcal{F}_x^\#$ is an isomorphism of abelian groups. This will be used below without further mention.

To prove the adjointness property we use the adjointness property of sheafification of presheaves of sets. For example if $\psi : \mathcal{F} \to i(\mathcal{G})$ is morphism of presheaves then we obtain a morphism of sheaves $\psi' : \mathcal{F}^\# \to \mathcal{G}$. What we have to do is to check that this is a morphism of abelian sheaves. We may do this for example by noting that it is true on stalks, by Lemma 007Z, and then using Lemma 007W above. $\square$